



PREDICTING RESIDENTIAL ELECTRICITY CONSUMPTION

Using CNN BiLSTM-SA Neural Networks

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Problem Statement



Electricity demand changes unpredictably.



Poor forecasting causes outages and inefficiency.



Traditional models show low accuracy.



A high-accuracy time series model is needed.



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Objective



Build a deep learning model for electricity prediction.



Improve accuracy using a hybrid architecture.



Compare performance with baseline models.

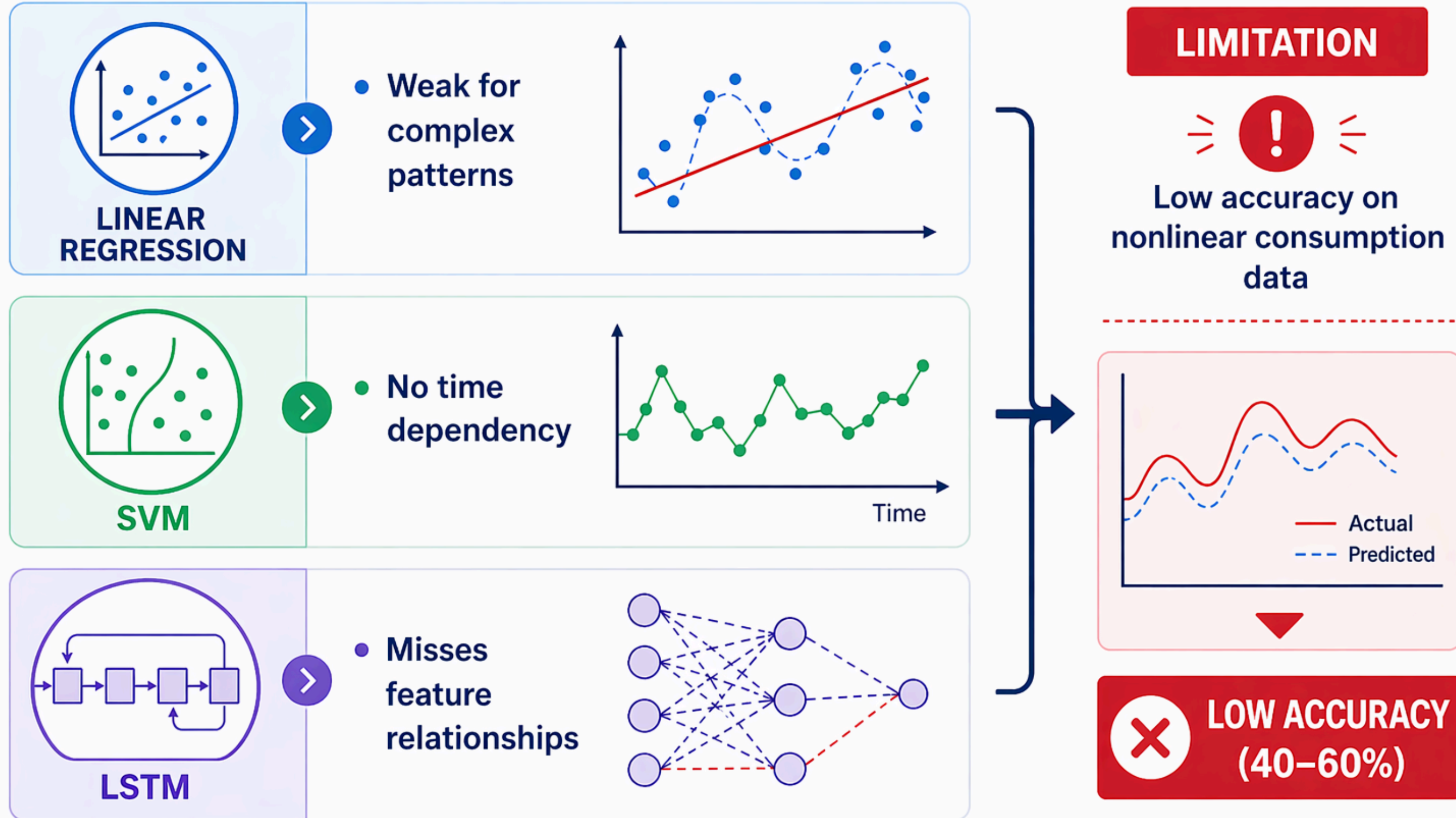


Reduce RMSE and MAE.



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Existing System



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Proposed System

CNN extracts local temporal features.

BiLSTM learns forward and backward dependencies.

Self-Attention highlights important features.

Output: more accurate electricity forecasting.



6 Dataset

Source :

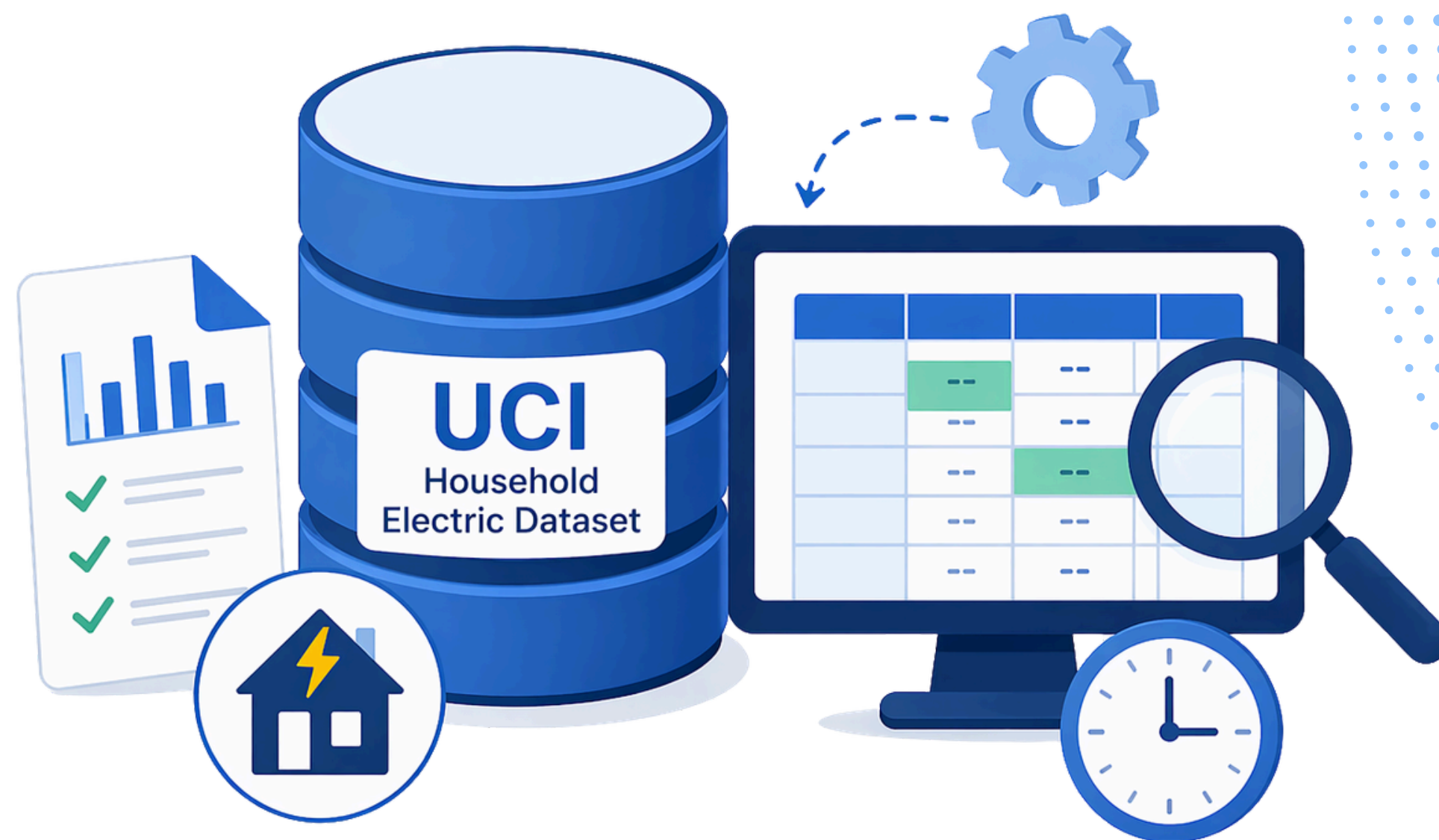
UCI Household Electric Dataset.

Features :

- Volatage
- Global Active Power
- Sub-metering values

Preprocessing :

- Missing value removal
- Normalization
- Time conversion.



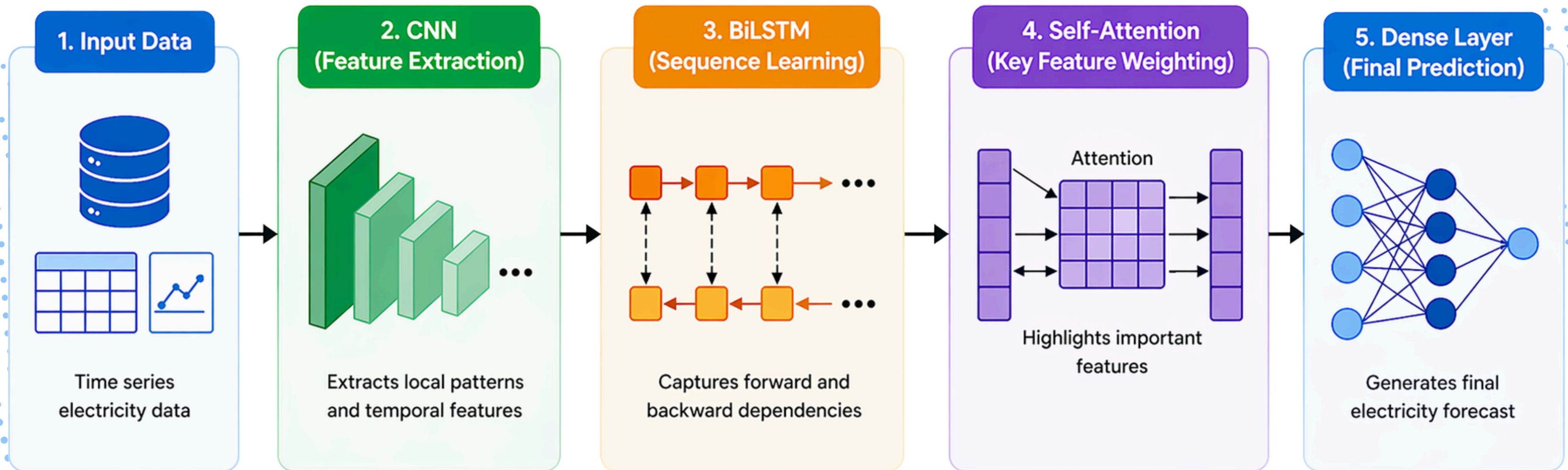
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Methodology (Pipeline)



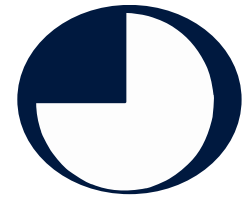
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Model Architecture



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Training Details



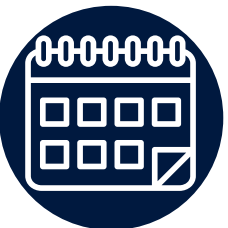
Train/Test split: 80/20



Loss function: MSE



Optimizer: Adam

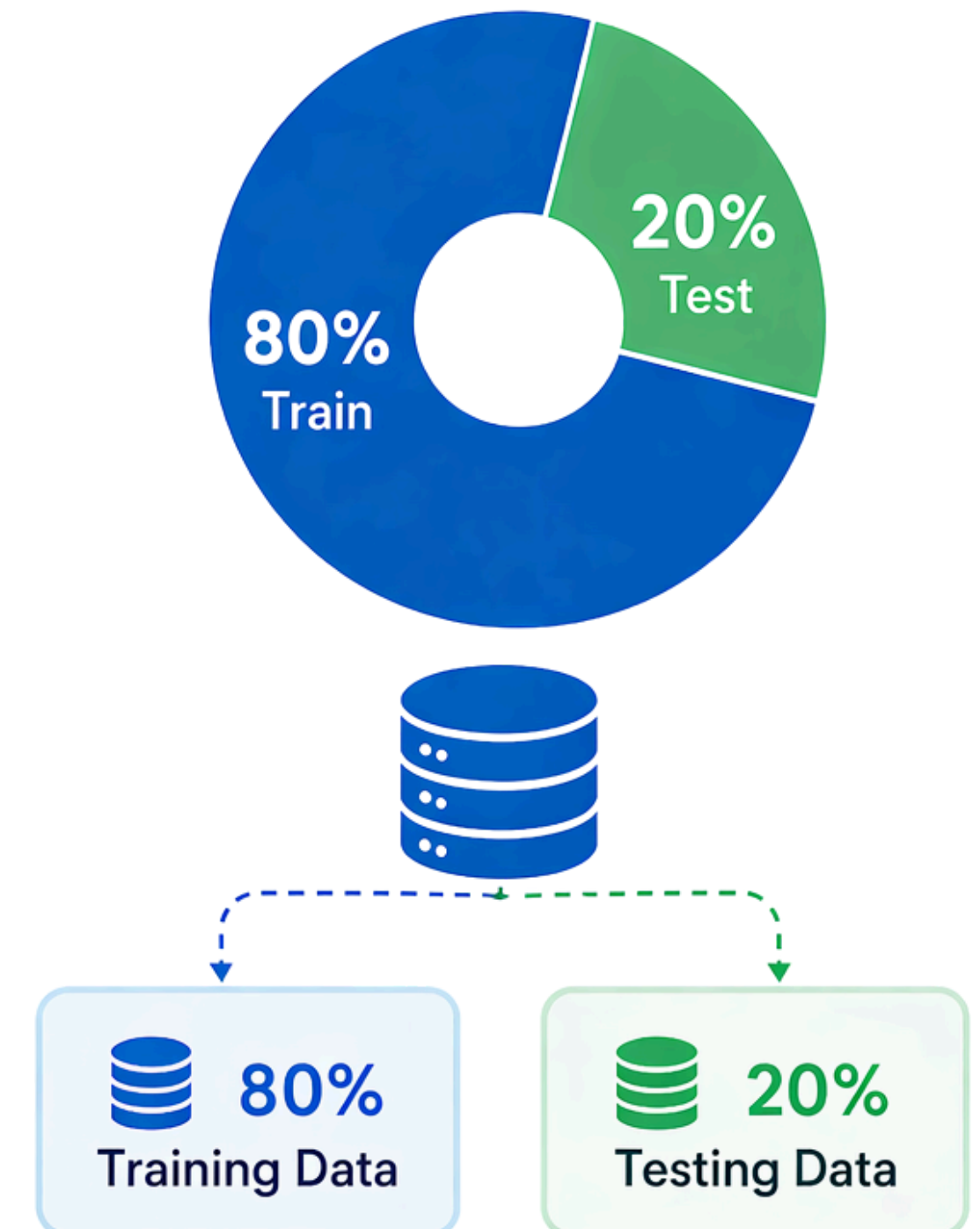


Epochs: use your actual code value



Batch size: use your actual code value

TRAIN/TEST SPLIT



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Evaluation Metrics



R^2 score: prediction accuracy



RMSE: error magnitude



MAE: average prediction error

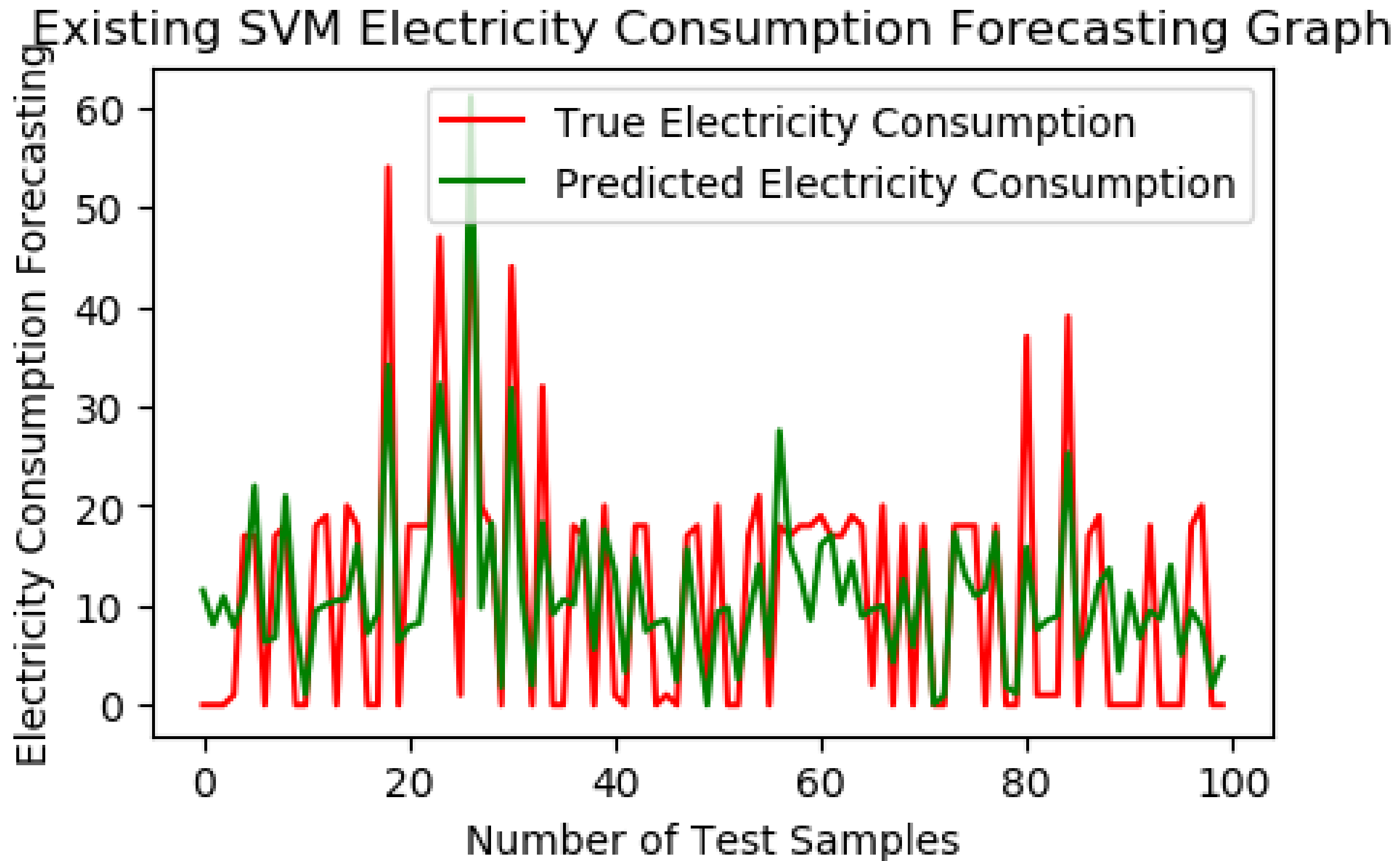


Goal: high R^2 , low RMSE and MAE



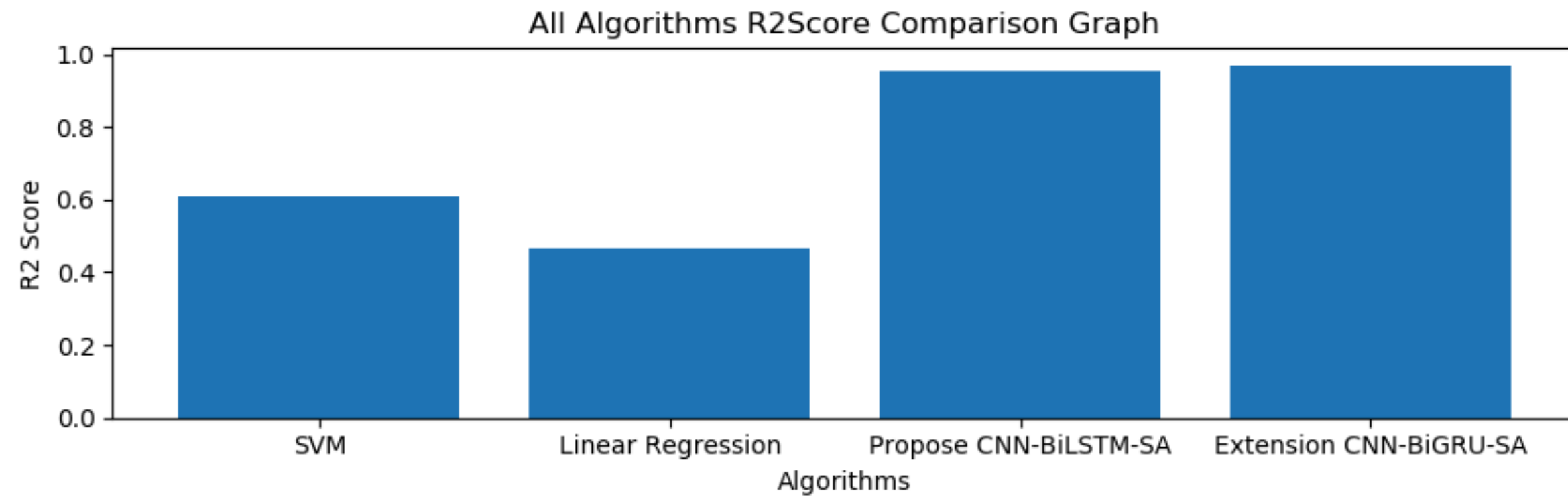
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Results (Actual vs Predicted)



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Model Comparison (R² Score)



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Result Analysis



Actual vs predicted values are plotted over time.



The predicted curve should closely follow the actual curve.



This shows strong forecasting performance.



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Extension Work



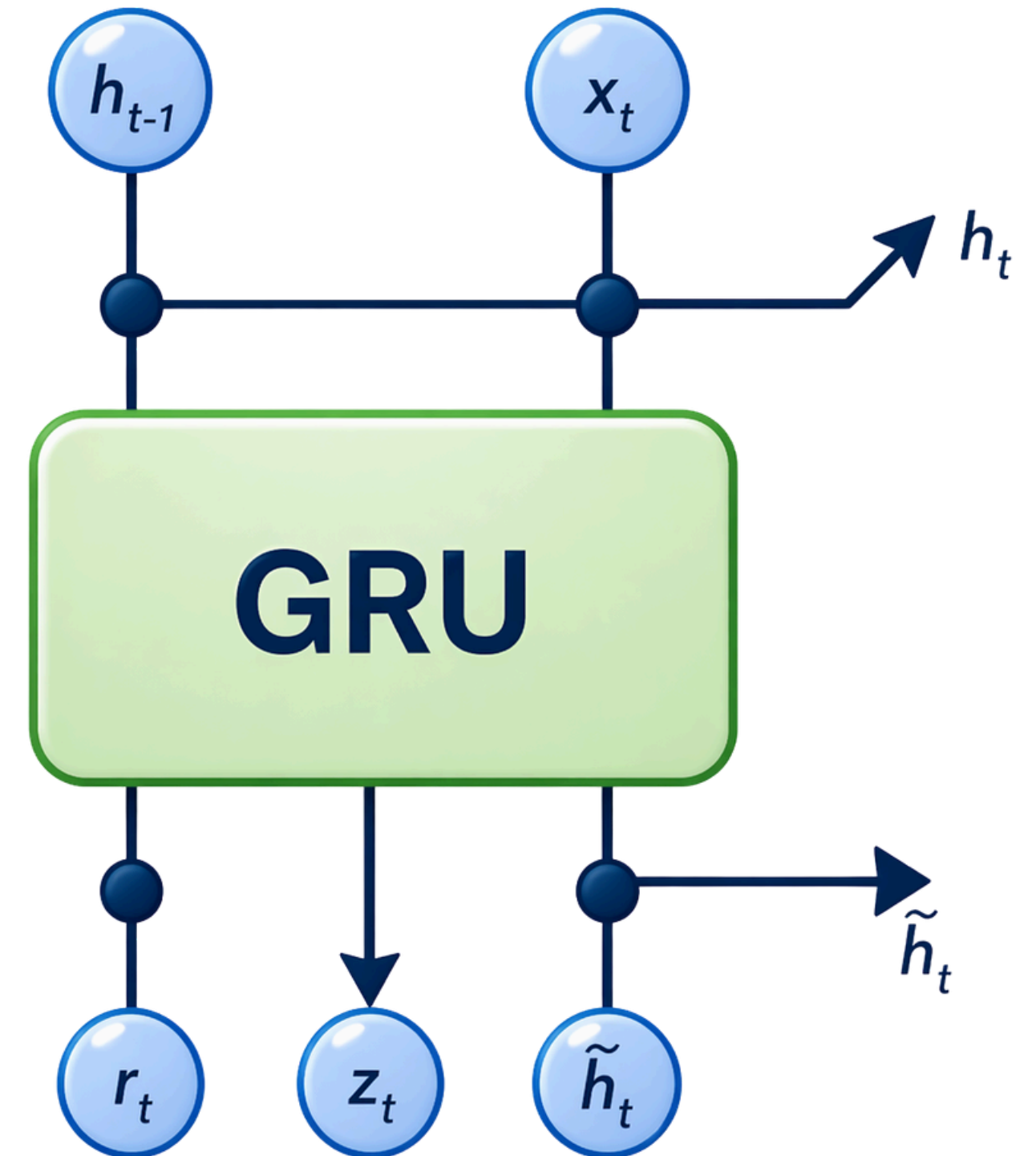
Replace BiLSTM with BiGRU.



Reduce training time.



Maintain or slightly improve accuracy.



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Advantages



High prediction accuracy



Better energy planning



Reduced outage risk



Scalable for smart grid use



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Conclusion



CNN + BiLSTM + Self-Attention outperforms traditional models.



The model is suitable for smart energy forecasting.



It improves both accuracy and reliability.



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Thank You

Thank,
you!

